



How to Run Multiple Instances of Firebird Server

Firebird is a relational database that runs on Linux, Windows and many UNIX platforms, and is commercially independent. It has been developed by C and C++ programmers in association with technical advisors and supporters to develop on the source code released by Borland Software Corp.

If it's the first time that you're reading about Firebird, then here's what you should know — Firebird technology has been in use for 20 years, which makes it a very mature and stable product. It's a fully featured RDBMS that can handle databases ranging in size from just a few KB to many gigabytes, with good performance, and it is almost maintenance-free. Another cool thing about Firebird is that the installer size usually is less than 7MB (depending on the OS) and it is fully automated. This might leave some of us thinking that Firebird can only be used with just small databases and a few connections. You're totally mistaken! It is used with many big databases and lots of connections.

Firebird comes with a full set of command line utilities that allow you to create databases, retrieve database statistics, run SQL commands and scripts, perform backups and restores, etc. If you prefer to use a GUI (graphical user interface) tool, there are lots of options to choose from.

The Firebird server comes in four flavours: SuperServer, Classic, SuperClassic and Embedded. If you are a newbie, I'd recommend you to start with SuperServer. Right now, Classic is recommended for use with SMP machines and some other specific situations. SuperServer shares its cache among the database connections, and it uses threads to handle each connection. It is a multi-threaded, multi-client architecture that requires few resources when additional connections are spawned off.

Resource allocation and lock management are much more efficient in the SuperServer architecture, because separate processes do not have to wait for others to finish before they can be addressed. Classic starts an independent server process for each connection made, while SuperClassic offers a multi-threaded single process server with an independent cache for each connection. One awesome feature about Firebird is that it allows you to run multiple servers on a single machine — for example, when you want to test a new version of Firebird or when you have different applications using different server versions and you want to use only one server machine, or if your testing/integration database is running on the same environment as your production database (this is usually not preferable, but you may require such a setup to cut costs).

Here's a quick guide on how that can be achieved in Firebird 3.0. The same procedure is applicable to other versions of Firebird as long as you want to use the Classic

Server. Mixing different versions should also work, because the required libraries will all be isolated in single directories.

Step 1: Installation and setup

Once you're done with installation, you will have two separate Firebird 3.0 instances in your machine. Let's call them firebird01 and firebird02, with two different `fb_inet_server` executables. To distinguish between them further, let's name their executables as `fb_inet_server01` and `fb_inet_server02` respectively. Firebird01 will listen on standard Firebird Port 3050 and firebird02 will listen on Port 3060, as shown below.

Name	Port	Directory	Executable
Firebird01	3050	/opt/firebird01	/opt/firebird01/bin/fb_inet_server01
Firebird02	3060	/opt/firebird02	/opt/firebird02/bin/fb_inet_server02

First, visit the official Firebird site and download the desired package. I chose *Firebird-3.0.4.33054-0.i686.tar.gz*.

We will need some dependencies such as `libstdc++5` in Ubuntu 10.10 Maverick. One can do `apt-get` to install it. Once you extract the package, you will find a file called *buildroot.tar.gz*, which holds all the files needed to get the Firebird 3.0 server up and running. After extraction, extract *FirebirdCS-3.0.4.33054-0.i686/buildroot.tar.gz* and copy the *FirebirdCS-3.0.4.33054-0.i686/opt/firebird* over to */opt/firebird01*.

```
root@host:/root# tar xzvf FirebirdCS-3.0.4.33054-0.i686.tar.gz
FirebirdCS-3.0.4.33054-0.i686/
FirebirdCS-3.0.4.33054-0.i686/scripts/
FirebirdCS-3.0.4.33054-0.i686/scripts/tarMainInstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/postuninstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/preinstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/preuninstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/taruninstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/postinstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/tarinstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/rpmfiles.txt
FirebirdCS-3.0.4.33054-0.i686/scripts/tarMainUninstall.sh
FirebirdCS-3.0.4.33054-0.i686/scripts/rpmheader.txt
FirebirdCS-3.0.4.33054-0.i686/install.sh
FirebirdCS-3.0.4.33054-0.i686/buildroot.tar.gz
FirebirdCS-3.0.4.33054-0.i686/manifest.txt
root@host:/root# cd FirebirdCS-3.0.4.33054-0.i686/
root@host:/root/FirebirdCS-3.0.4.33054-0.i686# tar xzvf
```